

Bootstrap 5 — Exercises

All 19 Modules — Complete Answers with Code and Output

1. Setting Up Bootstrap 5

Exercise 1.1 — Basic HTML Page with Bootstrap 5 CDN

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Bootstrap 5 CDN</title>
  <!-- Bootstrap 5 CSS via CDN -->
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
        rel="stylesheet" />
</head>
<body>
  <div class="container mt-4">
    <h1 class="text-primary">Hello, Bootstrap 5!</h1>
    <p class="lead">This page is linked to Bootstrap via CDN.</p>
    <button class="btn btn-success">Click Me</button>
  </div>
  <!-- Bootstrap 5 JS Bundle (Popper included) -->
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js">
</script>
</body>
</html>
```

Browser output:

```
Page title   : Bootstrap 5 CDN
Heading      : "Hello, Bootstrap 5!" styled in Bootstrap primary blue
Paragraph    : "This page is linked to Bootstrap via CDN." in larger lead font
Button       : Green "Click Me" button using btn btn-success
Layout       : Centred container with top margin
```

Exercise 1.2 — Project Using Downloaded Bootstrap Files

```
# Step 1: Download Bootstrap from https://getbootstrap.com/docs/5.3/getting-started/download/
# Step 2: Extract and place files in project folder:
#   project/
#     index.html
#     css/bootstrap.min.css
#     js/bootstrap.bundle.min.js

<!-- index.html using local files -->
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Bootstrap Local Files</title>
  <link rel="stylesheet" href="css/bootstrap.min.css" />
</head>
<body>
  <div class="container mt-5">
    <div class="alert alert-info">
      Bootstrap loaded from local downloaded files!
    </div>
  </div>
  <script src="js/bootstrap.bundle.min.js"></script>
</body>
```

```
</html>
```

Browser output:

Blue info alert box: "Bootstrap loaded from local downloaded files!"

No network request to CDN – styles load from local css/bootstrap.min.css

DevTools → Network: bootstrap.min.css loaded from local path (not cdn.jsdelivr.net)

2. Bootstrap Structure and Files

Exercise 2.1 — Explore Downloaded Bootstrap Directory

```
bootstrap/
  css/
    bootstrap.css          -- Full unminified CSS (development use)
    bootstrap.min.css      -- Minified CSS (production use)
    bootstrap.rtl.min.css  -- Right-to-left language support
    bootstrap-grid.min.css -- Grid system only (lighter alternative)
    bootstrap-reboot.min.css -- Reboot/normalize styles only
    bootstrap-utilities.min.css -- Utility classes only

  js/
    bootstrap.js          -- Full JS with comments (development)
    bootstrap.min.js      -- Minified JS without Popper (needs separate Popper)
    bootstrap.bundle.min.js -- Minified JS + Popper bundled (recommended)
    bootstrap.esm.min.js  -- ES Module version for modern bundlers

  icons/ (if Bootstrap Icons package downloaded separately)
    bootstrap-icons.css
  fonts/
    bootstrap-icons.woff
    bootstrap-icons.woff2

Folder summary:
css/ -- All stylesheet variants; use bootstrap.min.css for production
js/  -- All script variants; use bootstrap.bundle.min.js (includes Popper)
icons/ -- SVG icon font; requires separate download from icons.getbootstrap.com
```

Exercise 2.2 — Include bootstrap.bundle.min.js

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Bootstrap JS Plugins</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
        rel="stylesheet" />
</head>
<body>
  <div class="container mt-4">
    <h2>JS Plugins Demo</h2>
    <!-- Tooltip (requires JS) -->
    <button class="btn btn-secondary"
            data-bs-toggle="tooltip"
            title="I am a tooltip!">
      Hover me
    </button>
  </div>
  <!-- bootstrap.bundle.min.js includes Popper – required for tooltips/dropdowns -->
  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js">
  </script>
  <script>
    // Initialise all tooltips on the page
    const tooltips = document.querySelectorAll('[data-bs-toggle="tooltip"]');
    tooltips.forEach(el => new bootstrap.Tooltip(el));
  </script>
</body>
</html>
```

Browser output:

Hovering over "Hover me" button shows a dark tooltip: "I am a tooltip!"
Without bootstrap.bundle.min.js the tooltip does not appear.
DevTools Console: no errors – Popper is bundled inside bootstrap.bundle.min.js

3. Fundamentals of Responsive Grid Layout

Exercise 3.1 — Three Columns: Stack / Two / Three per Row

```
<div class="container mt-4">
  <div class="row">
    <!-- col-12: full width on mobile (stack)
         col-md-6: two per row on tablet
         col-lg-4: three per row on desktop -->
    <div class="col-12 col-md-6 col-lg-4 mb-3">
      <div class="p-3 bg-primary text-white text-center">Column 1</div>
    </div>
    <div class="col-12 col-md-6 col-lg-4 mb-3">
      <div class="p-3 bg-secondary text-white text-center">Column 2</div>
    </div>
    <div class="col-12 col-md-6 col-lg-4 mb-3">
      <div class="p-3 bg-success text-white text-center">Column 3</div>
    </div>
  </div>
</div>
```

Responsive behaviour:

Mobile (<768px) : Each column is full width – all three stack vertically
Tablet (>=768px) : Two columns per row (col-md-6 = 50% width each)
Desktop (>=992px) : Three columns per row (col-lg-4 = 33.3% width each)

Exercise 3.2 — container, row, and col-* Classes

```
<!-- .container      : fixed-width centred container with responsive breakpoints -->
<!-- .container-fluid: full-width container spanning 100% of viewport      -->
<!-- .row            : flex container, negative margins compensate column padding -->
<!-- .col-*          : columns must always be direct children of .row      -->

<div class="container">
  <div class="row g-3"> <!-- g-3 = gap/gutter between columns -->
    <div class="col-sm-6 col-md-4">
      <div class="border p-2 text-center">A</div>
    </div>
    <div class="col-sm-6 col-md-4">
      <div class="border p-2 text-center">B</div>
    </div>
    <div class="col-sm-12 col-md-4">
      <div class="border p-2 text-center">C</div>
    </div>
  </div>
</div>
```

Output:

Columns A, B, C render inside a centred container.
On small screens: A and B share a row; C spans full width.
On medium+: all three share one row equally (col-md-4 = 33.3% each).
g-3 adds consistent gaps between all columns.

4. Column Layouts and Grid Classes

Exercise 4.1 — Sidebar (col-md-3) and Content Area (col-md-9)

```
<div class="container mt-4">
  <div class="row">
    <!-- Sidebar: 3/12 columns on md+ screens -->
    <div class="col-12 col-md-3 bg-light p-3">
      <h5>Sidebar</h5>
      <ul class="list-unstyled">
        <li><a href="#">Home</a></li>
        <li><a href="#">About</a></li>
        <li><a href="#">Contact</a></li>
      </ul>
    </div>
    <!-- Main content: 9/12 columns on md+ screens -->
    <div class="col-12 col-md-9 p-3">
      <h2>Main Content Area</h2>
      <p>This area spans 9 of 12 columns on medium and larger screens.
        On mobile it stacks below the sidebar and takes full width.</p>
    </div>
  </div>
</div>
```

Output:

Desktop: Sidebar on left (25% width), Content on right (75% width)
Mobile : Sidebar stacks above content, both full width

Exercise 4.2 — Four Equal Columns (col-sm-3)

```
<div class="container mt-4">
  <div class="row">
    <div class="col-sm-3">
      <div class="card text-center p-2">Box 1</div>
    </div>
    <div class="col-sm-3">
      <div class="card text-center p-2">Box 2</div>
    </div>
    <div class="col-sm-3">
      <div class="card text-center p-2">Box 3</div>
    </div>
    <div class="col-sm-3">
      <div class="card text-center p-2">Box 4</div>
    </div>
  </div>
</div>
```

Output:

On sm+ screens (>=576px): four equal columns side by side (each 25% wide)
On xs (<576px): columns stack vertically, each full width
col-sm-3 = 3 out of 12 grid columns = 25% per column

5. Alignment and Reordering in Grid

Exercise 5.1 — justify-content-center and align-items-center

```
<div class="container mt-4">
  <!-- justify-content-center: centres columns horizontally in the row -->
  <!-- align-items-center: centres columns vertically within the row height -->
  <div class="row justify-content-center align-items-center"
    style="height: 200px; background: #f8f9fa; border: 1px solid #dee2e6;">
    <div class="col-4">
      <div class="p-3 bg-primary text-white text-center">Centred Column 1</div>
    </div>
    <div class="col-4">
      <div class="p-3 bg-secondary text-white text-center">Centred Column 2</div>
    </div>
  </div>
</div>
```

Output:

Two columns are centred horizontally – empty space on left and right sides.
Columns are also vertically centred within the 200px row height.
Remaining 4 unused columns (12 - 4 - 4 = 4) split equally on both sides.

Exercise 5.2 — Reorder Columns with order-md-* Classes

```
<div class="container mt-4">
  <div class="row">
    <!-- On mobile: renders in DOM order (A then B)
         On md+: order-md-2 moves A second, order-md-1 moves B first -->
    <div class="col-12 col-md-6 order-md-2">
      <div class="p-3 bg-warning text-center">
        Column A (DOM first, visually second on desktop)
      </div>
    </div>
    <div class="col-12 col-md-6 order-md-1">
      <div class="p-3 bg-info text-center">
        Column B (DOM second, visually first on desktop)
      </div>
    </div>
  </div>
</div>
```

Output:

Mobile : A appears above B (DOM order)
Desktop : B appears on left, A appears on right (visually reordered)
Useful for placing content before sidebar in DOM (SEO/accessibility)
while showing sidebar on the left visually on wide screens.

6. Responsive Flexbox Utilities

Exercise 6.1 — Responsive Navbar with d-flex and flex-md-row

```
<nav class="navbar bg-dark p-3">
  <!-- d-flex flex-column: stack links vertically on mobile -->
  <!-- flex-md-row: switch to horizontal on md+ screens -->
  <div class="d-flex flex-column flex-md-row align-items-start
    align-items-md-center gap-3">
    <a href="#" class="text-white text-decoration-none fw-bold">MyBrand</a>
    <a href="#" class="text-white text-decoration-none">Home</a>
    <a href="#" class="text-white text-decoration-none">About</a>
    <a href="#" class="text-white text-decoration-none">Contact</a>
  </div>
</nav>
```

Output:

Mobile : Links stack vertically in a column (flex-column)
Tablet+ : Links align horizontally in a row (flex-md-row)
gap-3 adds consistent spacing between each link in both orientations.

Exercise 6.2 — justify-content-between and align-items-center in Card

```
<div class="container mt-4">
  <div class="card p-3">
    <!-- d-flex + justify-content-between: pushes title left, button right -->
    <!-- align-items-center: vertically centres both on the same baseline -->
    <div class="d-flex justify-content-between align-items-center mb-3">
      <h5 class="mb-0">Card Title</h5>
      <button class="btn btn-sm btn-outline-primary">View All</button>
    </div>
    <p class="text-muted">Card content goes here with description text.</p>
  </div>
</div>
```

Output:

Card header row: "Card Title" aligned left, "View All" button aligned right.
Both are vertically centred on the same line regardless of font size.
justify-content-between distributes space so items go to opposite ends.

7. Typography

Exercise 7.1 — Bootstrap Typography Utilities

```
<div class="container mt-4">
  <p class="display-1">Display 1 – Hero Heading</p>
  <p class="display-4">Display 4 – Section Heading</p>
  <p class="lead">Lead paragraph – slightly larger text for introductions.</p>
  <p class="text-muted">Muted text – lighter grey colour for secondary info.</p>
  <p class="fw-bold">Bold text using fw-bold utility.</p>
  <p class="fw-light">Light weight text using fw-light.</p>
  <p class="fst-italic">Italic text using fst-italic.</p>
  <p class="text-decoration-underline">Underlined text.</p>
  <p class="fs-1">Font size 1 (largest via fs-* scale)</p>
  <p class="fs-6">Font size 6 (smallest via fs-* scale)</p>
  <blockquote class="blockquote">
    <p>A well-known quote goes here.</p>
    <footer class="blockquote-footer">Author Name</footer>
  </blockquote>
</div>
```

Output:

```
display-1      : Very large hero text (~96px)
display-4      : Large section heading (~54px)
lead           : 1.25rem font-size, 300 weight – good for intros
text-muted     : #6c757d grey – secondary information
fw-bold        : font-weight 700
fw-light       : font-weight 300
fst-italic     : font-style italic
fs-1 to fs-6   : Bootstrap's 6-step font-size scale
blockquote-footer: small grey citation text below the quote
```

Exercise 7.2 — text-uppercase, text-lowercase, text-capitalize

```
<div class="container mt-4">
  <p class="text-uppercase">this text becomes all uppercase</p>
  <p class="text-lowercase">THIS TEXT BECOMES ALL LOWERCASE</p>
  <p class="text-capitalize">every first letter is capitalised here</p>
  <p class="text-start">Left aligned text</p>
  <p class="text-center">Centre aligned text</p>
  <p class="text-end">Right aligned text</p>
</div>
```

Output:

```
text-uppercase  : "THIS TEXT BECOMES ALL UPPERCASE"
text-lowercase  : "this text becomes all lowercase"
text-capitalize : "Every First Letter Is Capitalised Here"
text-start/center/end: left / centre / right alignment respectively
(These are CSS text-transform utilities – original HTML text is unchanged)
```


8. Forms

Exercise 8.1 — Registration Form

```
<div class="container mt-4" style="max-width: 500px;">
  <h3>Register</h3>
  <form>
    <div class="mb-3">
      <label for="fullName" class="form-label">Full Name</label>
      <input type="text" class="form-control" id="fullName"
        placeholder="Enter your full name" required />
    </div>

    <div class="mb-3">
      <label for="email" class="form-label">Email address</label>
      <!-- input-group: combines icon/text and input in one bordered box -->
      <div class="input-group">
        <span class="input-group-text">@</span>
        <input type="email" class="form-control" id="email"
          placeholder="email@example.com" />
      </div>
    </div>

    <div class="mb-3">
      <label for="password" class="form-label">Password</label>
      <input type="password" class="form-control" id="password" />
    </div>

    <div class="mb-3">
      <label class="form-label">Gender</label>
      <div class="form-check">
        <input class="form-check-input" type="radio"
          name="gender" id="male" value="male" />
        <label class="form-check-label" for="male">Male</label>
      </div>
      <div class="form-check">
        <input class="form-check-input" type="radio"
          name="gender" id="female" value="female" />
        <label class="form-check-label" for="female">Female</label>
      </div>
    </div>

    <div class="mb-3 form-check">
      <input type="checkbox" class="form-check-input" id="terms" />
      <label class="form-check-label" for="terms">
        I agree to the Terms and Conditions
      </label>
    </div>

    <button type="submit" class="btn btn-primary w-100">Register</button>
  </form>
</div>
```

Output:

Full Name field with placeholder text
Email with @ prefix inside input-group border
Password field (text masked)
Radio buttons for gender (only one selectable at a time)
Checkbox for terms agreement
Full-width Register button in blue

Exercise 8.2 — Login Form with form-floating

```
<div class="container mt-4" style="max-width: 400px;">
  <h3>Login</h3>
  <form>
    <!-- form-floating: label floats above input when user types -->
    <div class="form-floating mb-3">
      <input type="email" class="form-control" id="loginEmail"
```

```
        placeholder="name@example.com" />
        <label for="loginEmail">Email address</label>
    </div>

    <div class="form-floating mb-3">
        <input type="password" class="form-control" id="loginPassword"
            placeholder="Password" />
        <label for="loginPassword">Password</label>
    </div>

    <div class="form-check mb-3">
        <input class="form-check-input" type="checkbox" id="rememberMe" />
        <label class="form-check-label" for="rememberMe">Remember me</label>
    </div>

    <button type="submit" class="btn btn-success w-100">Login</button>
</form>
</div>
```

Output:

Email and Password fields show label as placeholder initially.
When user clicks inside, label smoothly floats to the top of the input.
form-floating requires placeholder attribute on the input to work correctly.
"Remember me" checkbox below fields.
Green full-width Login button.

9. Buttons

Exercise 9.1 — All Contextual Button Classes

```
<div class="container mt-4">
  <h5>Solid Buttons</h5>
  <button class="btn btn-primary m-1">Primary</button>
  <button class="btn btn-secondary m-1">Secondary</button>
  <button class="btn btn-success m-1">Success</button>
  <button class="btn btn-danger m-1">Danger</button>
  <button class="btn btn-warning m-1">Warning</button>
  <button class="btn btn-info m-1">Info</button>
  <button class="btn btn-light m-1">Light</button>
  <button class="btn btn-dark m-1">Dark</button>
  <button class="btn btn-link m-1">Link</button>

  <h5 class="mt-3">Outline Buttons</h5>
  <button class="btn btn-outline-primary m-1">Primary</button>
  <button class="btn btn-outline-secondary m-1">Secondary</button>
  <button class="btn btn-outline-success m-1">Success</button>
  <button class="btn btn-outline-danger m-1">Danger</button>
  <button class="btn btn-outline-warning m-1">Warning</button>
  <button class="btn btn-outline-info m-1">Info</button>
  <button class="btn btn-outline-dark m-1">Dark</button>

  <h5 class="mt-3">Sizes</h5>
  <button class="btn btn-primary btn-lg">Large</button>
  <button class="btn btn-primary">Default</button>
  <button class="btn btn-primary btn-sm">Small</button>

  <h5 class="mt-3">Disabled State</h5>
  <button class="btn btn-primary" disabled>Disabled</button>
</div>
```

Output:

- Row 1: 9 solid colour buttons (blue, grey, green, red, yellow, cyan, white, black, plain link)
- Row 2: 7 outlined buttons (transparent fill, coloured border and text)
- Row 3: 3 buttons showing lg / default / sm size variants
- Row 4: Greyed-out disabled button (cursor: not-allowed)

Exercise 9.2 — Button Groups and Toggle Buttons

```
<div class="container mt-4">
  <!-- btn-group wraps buttons into a single unit with shared borders -->
  <h5>Button Group</h5>
  <div class="btn-group" role="group">
    <button type="button" class="btn btn-outline-primary">Left</button>
    <button type="button" class="btn btn-outline-primary">Middle</button>
    <button type="button" class="btn btn-outline-primary">Right</button>
  </div>

  <h5 class="mt-3">Toggle Checkboxes (checked = active style)</h5>
  <div class="btn-group" role="group">
    <input type="checkbox" class="btn-check" id="chk1" autocomplete="off" />
    <label class="btn btn-outline-success" for="chk1">Option A</label>

    <input type="checkbox" class="btn-check" id="chk2" autocomplete="off" />
    <label class="btn btn-outline-success" for="chk2">Option B</label>

    <input type="checkbox" class="btn-check" id="chk3" autocomplete="off" />
    <label class="btn btn-outline-success" for="chk3">Option C</label>
  </div>

  <h5 class="mt-3">Button Toolbar</h5>
  <div class="btn-toolbar gap-2">
    <div class="btn-group">
      <button class="btn btn-primary">1</button>
      <button class="btn btn-primary">2</button>
    </div>
  </div>
```

```
<div class="btn-group">
  <button class="btn btn-secondary">A</button>
  <button class="btn btn-secondary">B</button>
</div>
</div>
</div>
```

Output:

btn-group: Left/Middle/Right share borders – no gaps between them

Toggle checkboxes: clicking Option A fills it green (active); clicking again unfills it

Multiple options can be selected independently (checkbox, not radio)

btn-toolbar: groups separated with a small gap

10. Navbars and Navigation

Exercise 10.1 — Responsive Navbar with Logo, Links, and Search

```
<nav class="navbar navbar-expand-md navbar-dark bg-dark">
  <div class="container">
    <!-- Brand / Logo -->
    <a class="navbar-brand fw-bold" href="#">MyBrand</a>

    <!-- Hamburger toggler for mobile -->
    <button class="navbar-toggler" type="button"
      data-bs-toggle="collapse"
      data-bs-target="#mainNav">
      <span class="navbar-toggler-icon"></span>
    </button>

    <!-- Collapsible content -->
    <div class="collapse navbar-collapse" id="mainNav">
      <!-- Nav links -->
      <ul class="navbar-nav me-auto mb-2 mb-md-0">
        <li class="nav-item">
          <a class="nav-link active" href="#">Home</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">Courses</a>
        </li>
        <li class="nav-item">
          <a class="nav-link" href="#">About</a>
        </li>
      </ul>

      <!-- Search form aligned right -->
      <form class="d-flex gap-2">
        <input class="form-control form-control-sm" type="search"
          placeholder="Search..." />
        <button class="btn btn-outline-light btn-sm" type="submit">
          Search
        </button>
      </form>
    </div>
  </div>
</nav>
```

Output:

Desktop: Brand on left, nav links in centre, search form on right – all in one row

Mobile : Brand shown, hamburger icon appears; clicking it expands links and search vertically below the brand

Active link "Home" appears slightly brighter than others

Exercise 10.2 — nav-tabs and nav-pills

```
<div class="container mt-4">
  <!-- Tab Navigation -->
  <h5>nav-tabs</h5>
  <ul class="nav nav-tabs" id="myTabs">
    <li class="nav-item">
      <a class="nav-link active" data-bs-toggle="tab" href="#tab1">Tab 1</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" data-bs-toggle="tab" href="#tab2">Tab 2</a>
    </li>
    <li class="nav-item">
      <a class="nav-link" data-bs-toggle="tab" href="#tab3">Tab 3</a>
    </li>
  </ul>
  <div class="tab-content border border-top-0 p-3">
    <div class="tab-pane fade show active" id="tab1">
      Content for Tab 1.
    </div>
  </div>
```

```
    <div class="tab-pane fade" id="tab2">Content for Tab 2.</div>
    <div class="tab-pane fade" id="tab3">Content for Tab 3.</div>
</div>

<!-- Pills Navigation -->
<h5 class="mt-4">nav-pills</h5>
<ul class="nav nav-pills gap-2">
  <li class="nav-item">
    <a class="nav-link active" href="#">All</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Frontend</a>
  </li>
  <li class="nav-item">
    <a class="nav-link" href="#">Backend</a>
  </li>
</ul>
</div>
```

Output:

nav-tabs: Underline-style tabs with content panel below; active tab is white on top
nav-pills: Pill-shaped rounded buttons; active one filled with primary blue
Clicking a tab fades in its content; other content fades out

11. Cards and Media Objects

Exercise 11.1 — Profile Card

```
<div class="container mt-4" style="max-width: 300px;">
  <div class="card shadow-sm">
    <!-- card-img-top: image sits flush at the top with rounded top corners -->
    
    <div class="card-body text-center">
      <h5 class="card-title">Rahul Sharma</h5>
      <p class="card-text text-muted">Software Engineer</p>
      <p class="card-text">
        <small class="text-muted">Mumbai, India</small>
      </p>
      <a href="#" class="btn btn-primary btn-sm">View Profile</a>
      <a href="#" class="btn btn-outline-secondary btn-sm ms-1">Message</a>
    </div>
    <div class="card-footer text-center text-muted">
      <small>Member since 2021</small>
    </div>
  </div>
</div>
```

Output:

Card with placeholder image at top (card-img-top – top corners rounded)
Name "Rahul Sharma" as card title (bold, larger)
"Software Engineer" in muted grey below
"Mumbai, India" in small text
Two buttons side by side: "View Profile" (filled) and "Message" (outlined)
Footer row with muted "Member since 2021" text

Exercise 11.2 — Media Object Layout (Image Left of Content)

```
<!-- Bootstrap 5 removed .media class – use flexbox utilities instead -->
<div class="container mt-4">
  <div class="d-flex gap-3 mb-4">
    <!-- Image aligned to the left -->
    
    <!-- Content to the right of the image -->
    <div>
      <h5 class="mt-0 mb-1">Priya Nair</h5>
      <p class="mb-1 text-muted">Posted a new article</p>
      <p class="mb-0">Bootstrap 5 removes the .media component.
        We replicate the same layout using d-flex and gap utilities,
        which is more flexible and idiomatic Bootstrap 5.</p>
    </div>
  </div>

  <!-- Nested media object -->
  <div class="d-flex gap-3 ms-5">
    
    <div>
      <h6 class="mt-0 mb-1">Arjun Mehta</h6>
      <p class="mb-0 text-muted">Replied: Great article, very helpful!</p>
    </div>
  </div>
</div>
```

Output:

Avatar image on left (80x80), text block on right separated by gap-3
Nested reply indented with ms-5 (margin-start), smaller 60x60 avatar
flex-shrink-0 prevents the image from shrinking if text is long

12. Spacing Utilities

Exercise 12.1 — Margin and Padding Utilities

```
<!-- Margin classes: m-{size}, mt-{size}, mb-{size}, ms-{size}, me-{size}, mx, my -->
<!-- Padding classes: p-{size}, pt-{size}, pb-{size}, ps-{size}, pe-{size}, px, py -->
<!-- Sizes: 0=0, 1=0.25rem, 2=0.5rem, 3=1rem, 4=1.5rem, 5=3rem -->

<div class="container">
  <div class="bg-primary text-white m-3 p-2">
    m-3 (margin all sides 1rem) + p-2 (padding all sides 0.5rem)
  </div>

  <div class="bg-success text-white mt-4 p-3">
    mt-4 (margin top 1.5rem) + p-3 (padding 1rem)
  </div>

  <div class="bg-warning text-dark py-5 px-3">
    py-5 (padding top+bottom 3rem) + px-3 (padding left+right 1rem)
  </div>

  <div class="bg-info text-white mb-0 ms-5 me-5 p-2">
    mb-0 + ms-5 + me-5: no bottom margin, large left and right margins
  </div>
</div>
```

Output:

```
Blue box   : 1rem gap around it (m-3), 0.5rem inner space (p-2)
Green box  : 1.5rem gap above it (mt-4), 1rem inner space (p-3)
Yellow box : 3rem top and bottom padding makes it taller (py-5)
Cyan box   : 3rem left/right margins push it inward from container edges
```

Exercise 12.2 — Pricing Section with Spacing

```
<div class="container py-5">
  <h2 class="text-center mb-5">Pricing Plans</h2>
  <div class="row justify-content-center g-4">
    <div class="col-md-4">
      <div class="card text-center p-4 shadow-sm">
        <h4 class="mb-3">Basic</h4>
        <p class="display-5 fw-bold mb-1">$9<small class="fs-6">/mo</small></p>
        <p class="text-muted mb-4">For individuals getting started.</p>
        <ul class="list-unstyled mb-4">
          <li class="mb-2">5 Projects</li>
          <li class="mb-2">2 GB Storage</li>
          <li class="mb-2">Email Support</li>
        </ul>
        <a href="#" class="btn btn-outline-primary mt-auto">Get Started</a>
      </div>
    </div>
    <div class="col-md-4">
      <div class="card text-center p-4 shadow border-primary">
        <h4 class="mb-3">Pro</h4>
        <p class="display-5 fw-bold mb-1">$29<small class="fs-6">/mo</small></p>
        <p class="text-muted mb-4">For growing teams.</p>
        <ul class="list-unstyled mb-4">
          <li class="mb-2">Unlimited Projects</li>
          <li class="mb-2">20 GB Storage</li>
          <li class="mb-2">Priority Support</li>
        </ul>
        <a href="#" class="btn btn-primary mt-auto">Get Started</a>
      </div>
    </div>
  </div>
</div>
```

Output:

```
py-5 on container: 3rem top and bottom breathing room
mb-5 on heading:   big gap below the title before the cards
```


g-4 on row: consistent gutters between the two price cards
mb-3/mb-4 inside: controlled vertical rhythm between card elements
List items spaced with mb-2 – readable without being too spread out

13. Colors and Backgrounds

Exercise 13.1 — Dashboard with Contextual Background Classes

```
<div class="container mt-4">
  <h2 class="mb-4">Dashboard</h2>
  <div class="row g-3">
    <div class="col-md-3">
      <div class="bg-primary text-white p-4 rounded text-center">
        <h5>Users</h5><p class="fs-3 fw-bold mb-0">1,240</p>
      </div>
    </div>
    <div class="col-md-3">
      <div class="bg-success text-white p-4 rounded text-center">
        <h5>Revenue</h5><p class="fs-3 fw-bold mb-0">$52k</p>
      </div>
    </div>
    <div class="col-md-3">
      <div class="bg-warning text-dark p-4 rounded text-center">
        <h5>Pending</h5><p class="fs-3 fw-bold mb-0">34</p>
      </div>
    </div>
    <div class="col-md-3">
      <div class="bg-danger text-white p-4 rounded text-center">
        <h5>Errors</h5><p class="fs-3 fw-bold mb-0">7</p>
      </div>
    </div>
  </div>

  <!-- Text colour utilities -->
  <div class="mt-4">
    <p class="text-primary">text-primary</p>
    <p class="text-success">text-success</p>
    <p class="text-danger">text-danger</p>
    <p class="text-warning">text-warning</p>
    <p class="text-info">text-info</p>
    <p class="text-muted">text-muted</p>
    <p class="text-dark">text-dark</p>
  </div>
</div>
```

Output:

4 stat cards in a row: blue (Users), green (Revenue), yellow (Pending), red (Errors)
Each card has bold number and white/dark text depending on background contrast
Text colour samples shown below in their respective colours

Exercise 13.2 — bg-gradient with bg-dark and White Text

```
<div class="container mt-4">
  <!-- bg-gradient adds a subtle CSS gradient on top of the background colour -->
  <div class="bg-dark bg-gradient text-white p-5 rounded text-center mb-3">
    <h2>Dark Gradient Banner</h2>
    <p class="lead">bg-dark + bg-gradient gives a sleek dark-to-slightly-lighter effect.</p>
    <a href="#" class="btn btn-outline-light">Learn More</a>
  </div>

  <div class="bg-primary bg-gradient text-white p-4 rounded mb-3">
    <h4>Primary Gradient</h4>
    <p>bg-primary + bg-gradient</p>
  </div>

  <div class="bg-success bg-gradient text-white p-4 rounded">
    <h4>Success Gradient</h4>
    <p>bg-success + bg-gradient</p>
  </div>
</div>
```

Output:

Dark banner with subtle gradient (dark at bottom, slightly lighter at top)

White text and outline button are clearly readable on dark background
bg-gradient works with any bg-{colour} class – adds a linear-gradient overlay
Bootstrap 5 enables gradients via \$enable-gradients: true in Sass (off by default)
CDN version does NOT show gradient unless custom compiled – gradient is subtle

14. Display and Visibility

Exercise 14.1 — d-none, d-md-block, d-lg-flex

```
<div class="container mt-4">

  <!-- Visible only on mobile (hidden on md and above) -->
  <div class="d-block d-md-none bg-warning p-2 mb-2">
    Mobile only content (hidden on tablet+)
  </div>

  <!-- Hidden on mobile, visible as block on md+ -->
  <div class="d-none d-md-block bg-info text-white p-2 mb-2">
    Tablet and above only (hidden on mobile)
  </div>

  <!-- Hidden until lg, shown as flex on lg+ -->
  <div class="d-none d-lg-flex justify-content-between bg-success text-white p-2 mb-2">
    <span>Desktop flex item 1</span>
    <span>Desktop flex item 2</span>
    <span>Desktop flex item 3</span>
  </div>

  <!-- Always visible -->
  <div class="bg-secondary text-white p-2">
    Always visible on all screen sizes
  </div>

</div>
```

Responsive display behaviour:

Mobile (<768px) : Yellow "Mobile only" + grey "Always visible" shown
Tablet (>=768px) : Cyan "Tablet and above" + grey shown; yellow hidden
Desktop (>=992px) : Cyan + green flex row + grey shown; yellow hidden

d-{breakpoint}-{value} pattern:

d-none	= display: none
d-block	= display: block
d-md-block	= display: block at md and above
d-lg-flex	= display: flex at lg and above

Exercise 14.2 — Responsive Sidebar (Tablet and Above Only)

```
<div class="container-fluid mt-4">
  <div class="row">
    <!-- Sidebar: hidden on mobile, shown on md+ -->
    <div class="col-md-3 d-none d-md-block bg-light p-3 min-vh-100">
      <h5>Sidebar</h5>
      <ul class="list-group list-group-flush">
        <li class="list-group-item">Dashboard</li>
        <li class="list-group-item">Reports</li>
        <li class="list-group-item">Settings</li>
      </ul>
    </div>

    <!-- Main content: full width on mobile, 9 cols on md+ -->
    <div class="col-12 col-md-9 p-4">
      <h2>Main Content</h2>
      <p>On mobile, the sidebar is hidden and this area takes full width.
        On tablet and above, the sidebar appears on the left.</p>
    </div>
  </div>
</div>
```

Output:

Mobile : Only main content shown, full width – no sidebar
Tablet+ : Sidebar on left (3 cols) and content on right (9 cols)
min-vh-100 makes sidebar extend full viewport height

15. Borders, Shadows, and Rounded Corners

Exercise 15.1 — Border Utilities and rounded-circle on Image

```
<div class="container mt-4">
  <!-- rounded-circle on img: makes image perfectly circular -->
  

  <!-- Border utilities demo -->
  <div class="d-flex flex-wrap gap-3">
    <div class="border p-3">border (default thin)</div>
    <div class="border border-primary p-3">border-primary</div>
    <div class="border border-danger border-2 p-3">border-danger border-2</div>
    <div class="border border-success border-3 p-3">border-success border-3</div>
    <div class="border-top border-dark p-3">border-top only</div>
    <div class="border border-0 p-3 bg-light">border-0 (removed)</div>
    <div class="rounded p-3 border">rounded (small corners)</div>
    <div class="rounded-pill px-4 py-2 border">rounded-pill (capsule)</div>
    <div class="rounded-0 p-3 border">rounded-0 (sharp corners)</div>
  </div>
</div>
```

Output:

Image: circular crop with thick blue border (border-3 = 3px, rounded-circle)
Boxes showing border thickness: border (1px), border-2 (2px), border-3 (3px)
Border colours match Bootstrap contextual colour names
border-top: only a top border line, no other sides
rounded-pill: pill / capsule shape (very high border-radius)
rounded-0: completely sharp square corners

Exercise 15.2 — shadow, shadow-lg, and rounded-pill in a Card

```
<div class="container mt-4">
  <div class="row g-4">
    <div class="col-md-4">
      <div class="card shadow p-3 text-center">
        <p class="mb-1 fw-bold">shadow</p>
        <p class="text-muted small">Default shadow</p>
      </div>
    </div>
    <div class="col-md-4">
      <div class="card shadow-lg p-3 text-center">
        <p class="mb-1 fw-bold">shadow-lg</p>
        <p class="text-muted small">Large shadow – more depth</p>
      </div>
    </div>
    <div class="col-md-4">
      <div class="card shadow-sm p-3 text-center">
        <p class="mb-1 fw-bold">shadow-sm</p>
        <p class="text-muted small">Small subtle shadow</p>
      </div>
    </div>
  </div>

  <!-- rounded-pill on a card (unusual but valid) -->
  <div class="card shadow-lg rounded-pill px-5 py-4 mt-4 text-center"
    style="max-width: 400px; margin: auto;">
    <h5 class="mb-0">rounded-pill Card</h5>
    <p class="mb-0 text-muted small">Very rounded card shape</p>
  </div>
</div>
```

Output:

3 cards showing shadow depth: shadow-sm (subtle), shadow (medium), shadow-lg (deep)
shadow-lg lifts the card visually – good for highlighting featured content
rounded-pill card has exaggerated rounded ends – capsule-shaped card

16. Positioning Utilities

Exercise 16.1 — Fixed Footer with position-fixed bottom-0

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>Fixed Footer</title>
  <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/css/bootstrap.min.css"
    rel="stylesheet"/>
</head>
<body class="pb-5"> <!-- pb-5: prevents page content from hiding under footer -->

  <div class="container mt-4">
    <h2>Page Content</h2>
    <p>Scroll down to see the footer always stays at the bottom of the viewport.</p>
  </div>

  <!-- Fixed footer stays at bottom of viewport even when scrolling -->
  <footer class="position-fixed bottom-0 start-0 end-0
    bg-dark text-white text-center py-2">
    &copy; 2024 MyWebsite. All rights reserved.
  </footer>

  <script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js">
  </script>
</body>
</html>
```

Output:

Footer bar fixed at bottom of viewport – stays visible when page is scrolled
start-0 end-0: stretches full width of the screen
pb-5 on body: prevents page content from being hidden behind the fixed footer
Dark background, white text, centred copyright notice

Exercise 16.2 — position-relative and position-absolute for Badge Overlay

```
<div class="container mt-4">
  <!-- position-relative on parent lets absolute child be positioned relative to it -->
  <div class="position-relative d-inline-block">

    

    <!-- Badge overlaid at top-right corner of the image -->
    <span class="position-absolute top-0 end-0 translate-middle
      badge rounded-pill bg-danger">
      NEW
    </span>

  </div>

  <!-- Another example: notification count on an icon -->
  <div class="position-relative d-inline-block ms-5 mt-3">
    <button class="btn btn-outline-secondary">
      Notifications
    </button>
    <span class="position-absolute top-0 start-100 translate-middle
      badge rounded-pill bg-danger">
      5
    </span>
  </div>
</div>
```

Output:

Image with red "NEW" pill badge overlaid at the top-right corner
translate-middle shifts badge so its centre sits exactly on the corner

top-0 end-0: positions at top-right of the relative parent
Notification button with red circle badge "5" at top-right corner
Badge sits outside the button boundary due to translate-middle

17. Icons with Bootstrap Icons

Exercise 17.1 — Social Media Icons in Footer

```
<head>
  <!-- Bootstrap Icons CDN -->
  <link rel="stylesheet"
    href="https://cdn.jsdelivr.net/npm/bootstrap-icons@1.11.0/font/bootstrap-icons.css"/>
</head>

<footer class="bg-dark text-white py-4 mt-5">
  <div class="container text-center">
    <p class="mb-2">Follow us on social media</p>
    <div class="d-flex justify-content-center gap-3 fs-4">
      <a href="#" class="text-white text-decoration-none">
        <i class="bi bi-facebook"></i>
      </a>
      <a href="#" class="text-white text-decoration-none">
        <i class="bi bi-twitter-x"></i>
      </a>
      <a href="#" class="text-white text-decoration-none">
        <i class="bi bi-instagram"></i>
      </a>
      <a href="#" class="text-white text-decoration-none">
        <i class="bi bi-linkedin"></i>
      </a>
      <a href="#" class="text-white text-decoration-none">
        <i class="bi bi-youtube"></i>
      </a>
      <a href="#" class="text-white text-decoration-none">
        <i class="bi bi-github"></i>
      </a>
    </div>
    <p class="mt-3 text-muted small">&copy; 2024 MyWebsite</p>
  </div>
</footer>
```

Output:

Dark footer with white social media icons in a horizontal row
Icons: Facebook, X (Twitter), Instagram, LinkedIn, YouTube, GitHub
fs-4 makes icons 1.5rem tall – clearly visible
Hovering shows pointer cursor; white colour inherited from text-white

Exercise 17.2 — Icon-Only Buttons Using Bootstrap Icons

```
<div class="container mt-4">
  <h5>Icon-only buttons (replace text labels)</h5>
  <div class="d-flex gap-2 mb-3">
    <button class="btn btn-primary" title="Edit">
      <i class="bi bi-pencil-fill"></i>
    </button>
    <button class="btn btn-danger" title="Delete">
      <i class="bi bi-trash-fill"></i>
    </button>
    <button class="btn btn-success" title="Save">
      <i class="bi bi-check-circle-fill"></i>
    </button>
    <button class="btn btn-secondary" title="Download">
      <i class="bi bi-download"></i>
    </button>
    <button class="btn btn-outline-primary" title="Search">
      <i class="bi bi-search"></i>
    </button>
    <button class="btn btn-warning" title="Settings">
      <i class="bi bi-gear-fill"></i>
    </button>
  </div>

  <h5>Icon + Text buttons</h5>
```



```
<div class="d-flex gap-2">
  <button class="btn btn-primary">
    <i class="bi bi-plus-lg me-1"></i> Add New
  </button>
  <button class="btn btn-outline-danger">
    <i class="bi bi-trash me-1"></i> Delete
  </button>
  <button class="btn btn-success">
    <i class="bi bi-upload me-1"></i> Upload
  </button>
</div>
</div>
```

Output:

Row 1: Six icon-only buttons – pencil (blue), trash (red), check (green), download (grey), search (outlined), gear (yellow)
title attribute shows tooltip with action name on hover
Row 2: Buttons with icon + text side by side – me-1 adds small right margin between icon and text for visual spacing

18. Bootstrap 5 JavaScript Plugins

Exercise 18.1 — Modal Popup Triggered by a Button

```
<div class="container mt-4">
  <!-- Trigger button -->
  <button class="btn btn-primary" data-bs-toggle="modal"
    data-bs-target="#demoModal">
    Open Modal
  </button>
</div>

<!-- Modal markup (can be placed anywhere in the body) -->
<div class="modal fade" id="demoModal" tabindex="-1"
  aria-labelledby="demoModalLabel" aria-hidden="true">
  <div class="modal-dialog modal-dialog-centered">
    <div class="modal-content">

      <div class="modal-header">
        <h5 class="modal-title" id="demoModalLabel">Confirm Action</h5>
        <button type="button" class="btn-close"
          data-bs-dismiss="modal" aria-label="Close"></button>
      </div>

      <div class="modal-body">
        Are you sure you want to proceed with this action?
        This cannot be undone.
      </div>

      <div class="modal-footer">
        <button type="button" class="btn btn-secondary"
          data-bs-dismiss="modal">Cancel</button>
        <button type="button" class="btn btn-danger">Yes, Proceed</button>
      </div>

    </div>
  </div>
</div>

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js">
</script>
```

Output:

```
Clicking "Open Modal" fades in a centred modal overlay
Modal has: header with title and X close button
          body with confirmation message
          footer with "Cancel" (dismisses modal) and "Yes, Proceed" buttons
Clicking Cancel, X, or outside the modal closes and fades it out
modal-dialog-centered vertically centres the dialog in the viewport
```

Exercise 18.2 — Collapsible Accordion

```
<div class="container mt-4" style="max-width: 600px;">
  <h4>FAQ Accordion</h4>

  <div class="accordion" id="faqAccordion">

    <!-- Item 1 – open by default -->
    <div class="accordion-item">
      <h2 class="accordion-header">
        <button class="accordion-button" type="button"
          data-bs-toggle="collapse"
          data-bs-target="#faq1">
          What is Bootstrap 5?
        </button>
      </h2>
      <div id="faq1" class="accordion-collapse collapse show"
        data-bs-parent="#faqAccordion">
        <div class="accordion-body">
```

```

        Bootstrap 5 is a free, open-source CSS framework for building
        responsive, mobile-first websites.
    </div>
</div>
</div>

<!-- Item 2 -->
<div class="accordion-item">
    <h2 class="accordion-header">
        <button class="accordion-button collapsed" type="button"
            data-bs-toggle="collapse"
            data-bs-target="#faq2">
            Does Bootstrap 5 require jQuery?
        </button>
    </h2>
    <div id="faq2" class="accordion-collapse collapse"
        data-bs-parent="#faqAccordion">
        <div class="accordion-body">
            No. Bootstrap 5 dropped the jQuery dependency entirely.
            It uses vanilla JavaScript internally.
        </div>
    </div>
</div>

<!-- Item 3 -->
<div class="accordion-item">
    <h2 class="accordion-header">
        <button class="accordion-button collapsed" type="button"
            data-bs-toggle="collapse"
            data-bs-target="#faq3">
            What are Bootstrap Icons?
        </button>
    </h2>
    <div id="faq3" class="accordion-collapse collapse"
        data-bs-parent="#faqAccordion">
        <div class="accordion-body">
            Bootstrap Icons is a free, open-source icon library from the
            Bootstrap team with over 1800 icons, available as SVG or icon font.
        </div>
    </div>
</div>
</div>
</div>

```

Output:

FAQ accordion with 3 items; first item expanded by default (collapse show)
 Clicking a header expands it and collapses the previously open item
 data-bs-parent="#faqAccordion" enforces only one panel open at a time
 Smooth expand/collapse animation provided by Bootstrap JS
 Chevron arrow in header rotates when item is open

19. Customization with Sass

Exercise 19.1 — Set Up Bootstrap 5 with Sass via npm

```
# Step 1: Initialise project and install Bootstrap + Sass
mkdir bootstrap-sass-project && cd bootstrap-sass-project
npm init -y
npm install bootstrap@5.3.0
npm install --save-dev sass
```

```
# Project structure after setup:
#   bootstrap-sass-project/
#     node_modules/
#     scss/
#       custom.scss    ← our custom Sass entry file
#     css/              ← compiled output folder
#     index.html
#     package.json
```

```
Terminal output:
added 2 packages in 3s
bootstrap@5.3.0 and sass installed successfully
```

scss/custom.scss — Import and Override Bootstrap

```
// Step 1: Override Bootstrap variables BEFORE importing Bootstrap
// (Variables use !default – they only set if not already defined)

$primary:      #e63946; // custom red instead of Bootstrap blue
$secondary:    #457b9d; // custom slate blue
$border-radius: 0.75rem; // more rounded corners globally
$font-family-base: 'Segoe UI', sans-serif;

// Step 2: Import Bootstrap source Sass files
// This compiles Bootstrap with our overrides applied
@import "../node_modules/bootstrap/scss/bootstrap";

// Step 3: Add any additional custom styles below
.hero-banner {
  background: linear-gradient(135deg, $primary, darken($primary, 15%));
  color: white;
  padding: 5rem 2rem;
  text-align: center;
}
```

Exercise 19.2 — Customise Primary Color and Border Radius, Then Recompile

```
# Add compile script to package.json scripts section:
# "build-css": "sass scss/custom.scss css/custom.css --style=compressed"
```

```
# Compile Bootstrap with custom variables
npm run build-css
```

```
Terminal output:
> sass scss/custom.scss css/custom.css --style=compressed
Compiled scss/custom.scss to css/custom.css
```

Output: css/custom.css (minified – ~190KB)

Result in browser (index.html links css/custom.css):
btn-primary buttons are now RED (#e63946) instead of Bootstrap blue (#0d6efd)
All rounded corners increased – cards, inputs, buttons have 0.75rem radius
Font changed to Segoe UI across the entire site

index.html — Using the Custom Compiled CSS

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8"/>
```

```

<meta name="viewport" content="width=device-width, initial-scale=1.0"/>
<title>Custom Bootstrap</title>
<!-- Custom compiled CSS (Bootstrap + overrides) -->
<link rel="stylesheet" href="css/custom.css"/>
</head>
<body>
  <div class="hero-banner">
    <h1 class="display-4">Custom Bootstrap Theme</h1>
    <p class="lead">Primary colour is now red. Border radius is 0.75rem.</p>
    <a href="#" class="btn btn-primary btn-lg mt-3">Get Started</a>
    <a href="#" class="btn btn-outline-light btn-lg mt-3 ms-2">Learn More</a>
  </div>

  <div class="container mt-5">
    <div class="card p-4 mb-3">
      <h5>Card with custom border-radius</h5>
      <p class="text-muted">Rounded corners are larger than Bootstrap default.</p>
      <input type="text" class="form-control" placeholder="Custom rounded input"/>
    </div>
  </div>

  <script src="node_modules/bootstrap/dist/js/bootstrap.bundle.min.js"></script>
</body>
</html>

```

Final browser output:

Hero section: dark red gradient background, white text, large red "Get Started" button
 Card: noticeably more rounded corners (0.75rem vs Bootstrap default 0.375rem)
 Input field: matching larger border-radius
 All btn-primary elements across the site use the custom \$primary red colour
 No CDN dependency – entire Bootstrap compiled locally with custom variables